Changing the rate of reaction (1)



Chemical reactions take place at different speeds. Some reactions, such as explosions, take place very quickly, while others, like the rusting of a car, can take years. The rate of a reaction is a measure of how quickly a chemical reaction occurs.

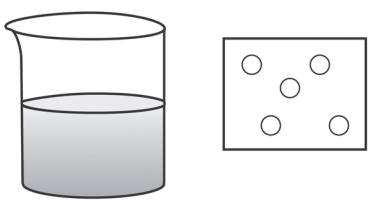
Reactions take place when particles collide with a certain amount of energy. Increasing the number of collisions and the energy that the particles collide with increases the rate of reaction. You can speed up or slow down a reaction by changing a number of different conditions.

1 Temperature

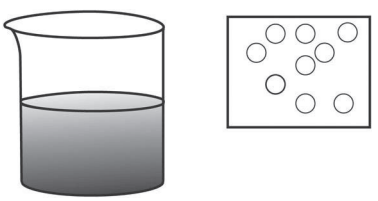
Increasing temperature speeds up the rate of reaction. As the temperature increases, the particles move faster and collide more often. This is why food is placed in fridges to slow down the chemical reactions that cause food to go off.

2 Concentration of solution

The concentration of a solution refers to the amount of a substance that is present in a certain amount of water. The more concentrated a solution is, the faster the reaction will occur, because there are more particles to collide into each other.



Less concentrated solution



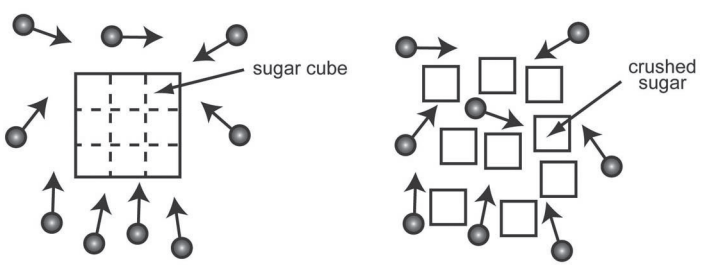
More concentrated solution

3 Size of particles

When a solid is crushed into smaller pieces there is more surface area in contact with the liquid. This means the particles collide with the liquid more often, speeding up the reaction.

4 Catalysts

Catalysts are substances that speed up the rate of reaction without being used up in the reaction.



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Changing the rate of reaction (2)

Read the Resource sheet 'Changing the rate of reaction (1)', then answer the following questions.

☞ **1** Use the words in the word bank to complete the passage.

Word bank
collide
surface area
catalysts
particles
rate
concentration
chemical
reaction
energy
temperature

Reactant p _____ must collide for a r_____ between them to take place. The r_____ of reaction increases when different reactant particles c_____ into each other more often and with more e_____ . The reaction rate can be sped up by increasing the s_____ a_____ of a solid, increasing the c_____ of a solution or increasing the t_____ . Chemicals called c_____ can also be used to speed up c_____ reactions.

2 Cross out the incorrect item from each pair in brackets.

As the temperature increases, the particles move (faster/slower) and collide (more/less) often.

The more concentrated a solution, the (faster/slower) the reaction rate.

When a solid is (crushed/whole), there is a bigger surface area for reactions to take place.

Catalysts are substances that (speed up/slow down) the rate of reaction and are (used up/not used up) during the reaction.

3 Are these statements true or false?

a Explosions are reactions that occur slowly. _____

b The rate of reaction is a measure of how quickly a chemical reaction occurs. _____

c Increasing the number of times and the energy with which reactant particles collide decreases the rate of reaction. _____

4 Rewrite the false statements to make them true.

_____

_____

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